

12 — Dataset Specification & Data Engineering Analysis

■ Overview of Project Datasets

The Maritime Port Intelligence Platform is powered by real-world, multi-year historical port traffic datasets stored in the `CODE_PH_0` directory.

```
data/ ■■■ port_cargo_monthly.csv # Primary NMPA Monthly Cargo Dataset (1,354 records, 2021–2026) ■■■
sample_cargo_data.csv # Multi-Port Weekly Benchmark Dataset (62 records, 2024)
```

■■ 1. Primary Dataset: `CODE_PH_0`

General Metadata

- **File Location:** `CODE_PH_0` (83.6 KB).
- **Domain Grounding:** Actual traffic records mapped to **New Mangalore Port Authority (NMPA)**.
- **Total Record Count:** 1,354 historical monthly observation rows.
- **Temporal Coverage:** Spans years **2021, 2023, 2024, 2025, and 2026**.
- `CODE_PH_0`: 72 unique commodity observations.
- `CODE_PH_0`: 22 commodity observations.
- `CODE_PH_0`: 16 commodity observations (Full multi-month coverage).
- `CODE_PH_0`: 16 commodity observations (Full multi-month coverage).
- `CODE_PH_0`: 22 commodity observations (Up to July 2026).

Column Schema & Data Definitions

Column Name	Data Type	Null Count	Description & Operational Meaning
<code>CODE_PH_0</code>	Object (str)	0	Name of the calendar month (e.g., <code>CODE_PH_0</code> , <code>CODE_PH_1</code> , <code>CODE_PH_2</code>).
<code>CODE_PH_0</code>	Int64	0	Calendar year (e.g., <code>CODE_PH_0</code> , <code>CODE_PH_1</code> , <code>CODE_PH_2</code> , <code>CODE_PH_3</code>).
<code>CODE_PH_0</code>	Object (str)	0	Trade flow direction: <code>CODE_PH_0</code> (Export) or <code>CODE_PH_1</code> (Import).

<i>CODE_PH_0</i>	Object (str)	0	Specific commodity category name (e.g., <i>CODE_PH_0</i> , <i>CODE_PH_1</i> , <i>CODE_PH_2</i>).
<i>CODE_PH_0</i>	Float64	0	Number of ship calls recorded for this commodity in current month.
<i>CODE_PH_0</i>	Float64	0	Number of ship calls for this commodity in the same month of previous year.
<i>CODE_PH_0</i>	Float64	0	Primary Metric: Total cargo weight handled (Metric Tonnes).
<i>CODE_PH_0</i>	Float64	0	Total cargo weight handled in same month of previous year (Tonnes).
<i>CODE_PH_0</i>	Float64	0	Percentage variation in tonnage volume YoY $\$((\text{Current} - \text{Prev})/\text{Prev}) \times 100\%$.

Commodity Categories Represented in NMPA Data

- CODE_PH_0*: Crude oil imports for MRPL (Mangalore Refinery & Petrochemicals Ltd).
- CODE_PH_0*: Strategic Crude Oil Reserves for Indian Strategic Petroleum Reserves Ltd.
- CODE_PH_0*: Thermal coal imports for UPCL (Udupi Power Corp Ltd / Adani Power Plant).
- CODE_PH_0*: Iron ore pellets and fines for KIOCL Bulk Terminal & Kudremukh Corridor.
- CODE_PH_0*: Containerized cargo handled at dedicated Berth 14 (JSW Container Terminal).
- CODE_PH_0*: Finished fertilizer imports for MCF (Mangalore Chemicals & Fertilizers Ltd).
- CODE_PH_0*: Fertilizer Raw Materials (Rock Phosphate, Sulphur).
- CODE_PH_0*: Liquefied Petroleum Gas imports for HPCL / BPCL / IOCL bottling plants.
- CODE_PH_0*: Petroleum, Oil & Lubricants (Clean products like Diesel, Petrol, Naphtha).
- CODE_PH_0*: Chemicals, Acids, Molasses, Bio-fuels.
- CODE_PH_0*: Palm oil and crude sunflower oil imports.
- CODE_PH_0*: Bulk cement and clinker shipments.
- CODE_PH_0*: Wheat, Rice, and Agricultural exports.
- CODE_PH_0*: Polypropylene, Paraxylene, Aromatic compounds.
- CODE_PH_0*: General dry bulk and break-bulk items.

■ Multi-Year Combination & Comparison Methodology

How does the system combine and compare **2024, 2025, and 2026** commodity datasets?

1. Continuous Time-Series Construction (*CODE_PH_0*):

- Converts *CODE_PH_0* and *CODE_PH_1* into standard datetime objects:

CODE_PH_0

- Sorts all 1,354 records chronologically from *CODE_PH_0* through *CODE_PH_1*.

2. Handling Missing Month-Commodity Gaps:

- If a specific commodity (e.g., *CODE_PH_0*) had zero shipments in a given month, the preprocessing pipeline performs a reindex grid search and fills missing volume values with *CODE_PH_1* tonnes rather than dropping rows.

3. Year-over-Year (YoY) Growth Calculations:

- Compares 2024 vs 2025 and 2025 vs 2026 tonnage by aligning identical calendar months:

$$\text{Growth \%} = \frac{\text{Volume}_{2026, \text{Month}} - \text{Volume}_{2025, \text{Month}}}{\text{Volume}_{2025, \text{Month}}} \times 100$$

4. Multi-Year Feature Engineering for ML:

- **Lag 1 Feature ($\$y_{t-1}$):** Volume in the immediately preceding month.
- **Lag 12 Feature ($\$y_{t-12}$):** Volume in the exact same month of the prior calendar year (captures annual seasonal cycles such as monsoon delays during June–August).

■ ■ 2. Secondary Benchmark Dataset: *CODE_PH_0*

- **File Location:** *CODE_PH_0* (2.6 KB).
- **Total Record Count:** 62 weekly records.
- **Fields:** *CODE_PH_0*, *CODE_PH_1* (Mumbai, Chennai, JNPT, Kandla, Vizag, Kochi), *CODE_PH_2*, *CODE_PH_3*, *CODE_PH_4*, *CODE_PH_5*, *CODE_PH_6*.
- **Usage:** Provides national benchmark metrics for comparing NMPA port performance against major Indian ports.